

Chapter 9 Homework: Matrices

Part A - Python Skills

1. Create two 2x3 matrices. 2. Perform addition. 3. Perform subtraction. 4. Perform Hadamard multiplication. 5. Perform element-wise division. 6. Create a 3x2 matrix and multiply it by a 2x2 matrix. 7. Multiply a matrix by a vector. 8. Multiply a matrix by a scalar. 9. Print matrix shapes before every operation.

Part B - Practical Problem

Suppose three students take two exams. Scores: $A = \begin{bmatrix} 85 & 90 \\ 78 & 88 \\ 92 & 95 \end{bmatrix}$ The instructor decides the exams have weights: $w = \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix}$ Tasks: 1. Use matrix-vector multiplication to compute each student's weighted score. 2. Explain why matrix multiplication is a natural representation of weighted averaging. 3. Add a third exam and modify the matrices appropriately. 4. Interpret the dimensions of every matrix and vector involved. 5. Explain how this resembles the matrix form of linear regression discussed in class.